

Original papers

Boosting crop disease recognition via automated image description generation and multimodal fusion

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ABSTRACT

In phytopathological diagnostics, traditional unimodal, vision-based methodologies often encounter limitations in performance due to the ambiguous manifestation of disease symptoms and the complexity of background environments. In particular, the robustness and generalizability of such methods are further limited under field conditions characterized by variable illumination, occlusion, and background clutter. Conversely, textual modality exhibits robust semantic encoding capabilities, enabling precise characterization of disease phenotypes and enhancing the discriminative capability of the model. However, textual modality remains underexploited and insufficiently integrated within current research paradigms. In this study, we propose a multimodal diagnostic framework that leverages large multimodal models to automatically generate structured textual descriptions from crop imagery, thereby reducing reliance on manual annotations. To further enhance cross-modal integration, a Projected Visual-Textual Discriminant (PVD) module is introduced. Empirical results indicate that by incorporating generated textual data, most multimodal architectures consistently outperform their unimodal equivalents across diverse visual backbone networks. Notably, the combination of CogAgent with CLIP (ViT-L/14) and the PVD module achieves an F1 score of 70.76%, while LLaVA combined with ResNet50+LSTM achieves 66.38%. The proposed methodology offers a practical and scalable solution for in-field crop disease diagnosis by better aligning visual and textual representations, enhancing classification performance, and obviating the need for manually curated textual annotations by automatically generating descriptions using the Automated Image Description Generation module.

1. Introduction

Limits of unimodal

Computer vision technology is increasingly employed in phytopathological diagnostics. In particular, deep learning-based unimodal models have greatly improved detection accuracy and efficiency. Convolutional neural networks (CNNs), such as VGG, ResNet, and DenseNet, have achieved remarkable success in foliar lesion recognition (Mehta et al., 2023; El Sakka et al., 2024). Lightweight variants such as MobileNet support real-time inference on resource-constrained devices (Karim et al., 2024). Vision Transformers (ViTs) further advance performance by enabling the modeling of long-range spatial dependencies (Doso-vitskiy, 2020; Xie and Ye, 2024). Despite these advances, unimodal, image-based systems remain vulnerable to environmental variability (illumination changes, occlusion, complex backgrounds, etc.) and similarity in symptoms. Such factors continue to undermine classification robustness (Ahad et al., 2023).

Unimodal methods also face additional challenges in real-field environments. For example, illumination changes exceeding 30% can reduce accuracy by > 25% and heavy occlusion markedly increases false positives (Barbedo, 2018; Too et al., 2019). Additionally, complex backgrounds and visually similar disease symptoms further complicate classification (Saleem et al., 2019). Moreover, unimodal systems lack biological and temporal context, which limits interpretability and prevents accurate severity assessment (Kamilaris and Prenafeta-Boldú, 2018). These shortcomings highlight the need for multimodal data fusion strategies to achieve robust performance in practical agricultural settings.

Based on these findings, researchers in different domains have increasingly adopted multimodal fusion techniques. In the biomedical field, multimodal fusion techniques have considerably improved diagnostic precision. For instance, Li et al. (2021) proposed a multimodal medical image fusion algorithm and Ghosh et al. (2024) combined

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0168-1699/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Contrastive Language–Image Pretraining (CLIP) with large multimodal models (LMMs) for medical image analysis. Reversible neural networks have also been utilized to advance clinical decision support (He et al., 2025). In addition, similar strategies have been explored in education and aerospace domains, confirming the broad generalizability of multimodal deep learning (Bewersdorff et al., 2025; Qian et al., 2024). These developments underscore the potential of multimodal techniques and their relevance to precision agriculture.

Multimodal learning has also shown promise in the domain of agriculture. The integration of visual data with sensor signals, temporal sequences, or expert annotations has yielded superior diagnostic accuracy compared to that achieved using unimodal models (El Sakka et al., 2025; Nabaei et al., 2025; Yuan et al., 2025; Yue et al., 2024). In addition, systems such as Anyplant for phenotypic monitoring (Nabaei et al., 2025) and edge-attention segmentation for fruit morphology (Wang et al., 2025) have demonstrated the value of fusing diverse inputs. In particular, image–text fusion has proven to be effective. Zhou et al. (2021) achieved > 99% accuracy by embedding disease images with textual descriptions. Transformer-based architectures further enhance flexibility and efficiency (Lu et al., 2024). Other systems, such as Rice-Fusion (Patil and Kumar, 2022), Tomato fusion networks (Zhang et al., 2022), and RustQNet (Deng et al., 2024), integrate either environmental or meteorological data with imagery, providing resilient multimodal fusion for robust disease diagnostics. Despite these advances, most existing approaches still rely on manually annotated image–text datasets. Such datasets are costly and scarce in agriculture. Direct application of generic multimodal models, such as CLIP or Bootstrapping Language–Image Pre-training (BLIP), often produces domain-specific misalignment. Fusion strategies can also become unstable under modality imbalance or complex field backgrounds (Radford et al., 2021). These challenges necessitate domain-adaptive frameworks that can automatically generate and align multimodal data.

Image captioning provides a promising solution by generating semantically rich descriptions from images (Vinyals et al., 2015; Ghandi et al., 2023; Xu et al., 2023). Transformer-based models such as BLIP and Flamingo (Karthik et al., 2024) can achieve strong performance in general domains. However, they struggle with agricultural terminology and temporal symptom progression. Recently, it has been demonstrated that LMMs, including LLaVA (Liu et al., 2024) and CogAgent (Hong et al., 2024), combined with structured prompting, can generate domain-specific descriptions that complement visual features.

Based on these insights, this study proposes an automated framework that converts unimodal image data into enriched multimodal resources. The framework generates descriptive captions for crop disease images and fuses them with visual features via adaptive mechanisms. A discriminative projection module, such as the Projected Visual–Textual Discriminant (PVD) (Wang et al., 2024), is used to enhance cross-modal alignment and classification performance. Unlike previous approaches that depend on costly manual annotations, the proposed method enables efficient and scalable multimodal learning in agriculture. This study lays the foundation for the practical deployment of multimodal crop disease diagnostics in real-world field conditions.

Key contributions of this study are as follows:

- This study introduces an automated text generation pipeline that applies large-scale multimodal models with structured Zero-shot Chain-of-Thought (CoT) prompts. The pipeline starts from image-only datasets and produces detailed descriptions of phytopathological images. These descriptions enable the construction of high-fidelity image–text pairs. Accordingly, the method reduces dependence on manual annotations and enables scalable multimodal classification in agricultural settings.
- This study develops a multimodal recognition architecture that integrates visual features with automatically generated text. We analyze fusion adaptation mechanisms between text generation

models (LLaVA and CogAgent) and visual feature extractors (ResNet50+LSTM and CLIP+PVD). To the best of our knowledge, this is the first comprehensive effort to jointly optimize automated text synthesis and multimodal fusion strategies for plant disease diagnostics.

- This study conducts extensive experiments on the PlantDoc dataset. The results show that CogAgent combined with CLIP (ViT-L/14) and PVD achieves an F1 score of 70.76%. LLaVA integrated with ResNet50 and LSTM achieves an F1 score of 66.38%. These findings confirm that our proposed automated pipeline delivers highly accurate multimodal classifications without relying on manually curated image descriptions.

2. Materials and methods

2.1. Overall architecture

Using image-only samples, this study proposes a multimodal learning framework that enhances diagnosis performance. The framework uses Automated Image Description Generation (AIDG) to generate textual inputs and fuses them with visual modalities. To the best of our knowledge, this is the first approach that augments image-only crop disease samples with automatically generated textual descriptions. Accordingly, the modality gap is bridged through a fully automated process. The design enables the system to benefit from multimodal advantages even when only image inputs are available. This reduces the dependency on manually written descriptions. This section provides a detailed description of the methodologies employed. First, the overall architecture of the proposed system is introduced to illustrate the general workflow. Subsequently, this study describes the automated text generation strategies based on LMMs. These include both the Zero-Shot CoT prompting strategy and a crop-specific prompting strategy designed to enhance semantic precision. Thereafter, two multimodal classification pipelines tailored for different application scenarios are presented. One is a ResNet50+LSTM fusion pipeline aimed at real-time inference. The other is a high-performance CLIP (ViT-L/14) pipeline enhanced with a PVD module to maximize recognition robustness. Both pipelines share the core idea of transforming image-only inputs into rich multimodal representations via automatic textual augmentation. Collectively, these components form a complete multimodal pipeline that starts from image-only data and automatically constructs paired inputs via AIDG.

As illustrated in Fig. 1, the framework first applies a large vision-language model (e.g., LLaVA-v1.6) with a Zero-Shot CoT prompting strategy. This model automatically generates structured textual descriptions of disease images. The original images are processed by a visual encoder, such as ResNet50 (He et al., 2016) or CLIP-ViT. The generated texts are embedded and encoded using a Long Short-Term Memory (LSTM) module or a CLIP-based text encoder. Multimodal features from both modalities are fused at the representation level. Fusion is performed via concatenation or projection-based alignment modules, such as PVD. The fused features are then passed through fully connected layers for classification. Optional dropout is applied to reduce overfitting. Finally, the model is optimized using a cross-entropy loss function to directly predict disease categories, without requiring manually written image descriptions.

2.2. Prompt engineering and template selection

This study constructed a structured template based on key diagnostic features in agricultural disease recognition and prior literature to generate high-quality image descriptions. The template covers visual attributes, such as leaf color, shape, texture, and spot distribution. This template was then used as the input prompt to guide LMMs (e.g., LLaVA) in description generation. Accordingly, the process ensures both semantic diversity and agricultural relevance.

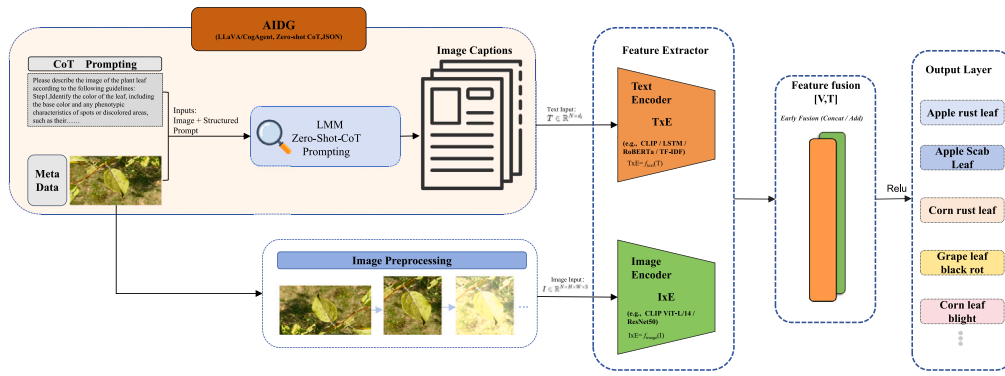


Fig. 1. Overall architecture of the proposed multimodal framework for crop disease diagnosis. The PVD module is only used in the CLIP-based framework, as indicated in the Fig. 4.

This approach is consistent with prior research that shows domain-specific prompt engineering enhances multimodal model performance (Gandhi et al., 2023; Sahoo et al., 2024). The generated descriptions were evaluated via random sampling to assess the quality. They were further refined iteratively with feedback from large models (e.g., GPT-4). This process combines expert-driven template design with the language generation capabilities of large models, balancing domain expertise with generative efficiency.

2.3. Automated image description generation using LMMs

Generate description

Given that the available data comprise image-only inputs, this study proposes an effective AIDG method based on the zero-sample CoT cueing strategy. The method uses an LMM to automatically generate textual descriptions of images. As depicted in Fig. 2, the model can produce multi-sentence natural language descriptions for input images without requiring manually written descriptions. The generated text captures key visual characteristics, such as leaf coloration, lesion distribution, morphological traits, and symptoms, including wilting and dryness. A structured Zero-Shot CoT prompting template is employed to ensure consistency and interpretability. This template guides the model through a step-by-step reasoning process aligned with crop disease diagnostic logic.

The entire pipeline operates in a fully automated, zero-shot setting without human supervision. For each image, a corresponding textual description is generated and compiled into a structured image–text pair. These pairs are stored in JSON format to ensure semantic alignment and support downstream multimodal training. This method augments image-only samples with AIDG-generated descriptions to reduce data preparation costs and expand the practical scope of multimodal learning in agriculture. It introduces a new paradigm for developing multimodal disease recognition systems from unimodal image sources. The approach offers scalability and automation for real-world applications.

Compared with traditional manually written descriptions, the proposed method achieves substantial efficiency gains. It can process ~1200 images per hour on a single A100 GPU, which is ~58 times faster than manual labeling. The manual evaluation of randomly selected samples confirms that the generated descriptions effectively capture disease-related visual features. These descriptions provide detailed semantic cues that are conducive to contrastive learning and classification tasks.

Formally, given an input image $I_i \in \mathcal{D}_{\text{img}}$, the corresponding structured textual description T_i is generated by a CoT-guided large multimodal model $\mathcal{G}_{\text{CoT}}$ as

$$T_i = \mathcal{G}_{\text{CoT}}(I_i). \quad (1)$$

The resulting image–text pairs form a pseudo-labeled dataset, which is expressed as follows

$$\mathcal{D}_{\text{pair}} = \{(I_i, T_i)\}_{i=1}^N, \quad (2)$$

where $\mathcal{D}_{\text{img}}$ denotes the original image dataset, $\mathcal{G}_{\text{CoT}}$ represents the image-to-text generation function guided by CoT prompting, and $\mathcal{D}_{\text{pair}}$ represents the generated image–text pairs used for downstream multimodal classification.

The automated generation of structured image–text pairs lays a solid foundation for improving multimodal crop disease recognition. This method substantially reduces data preparation costs by eliminating the need for manual annotation. Moreover, it expands the practical scope of multimodal learning in agriculture. The approach provides an efficient and scalable method to develop high-quality datasets. These datasets support the development of robust and deployable diagnostic systems. Automated description generation serves as the first stage of a multimodal pipeline. The pipeline begins with raw image inputs and enables scalable classification.

2.4. Feature extractor

The feature extraction module comprises two components: text feature extraction (TxE) and image feature extraction (IxE). The textual features are derived from automatically generated descriptions of the input images. Two distinct processing pipelines are designed: one uses ResNet50 for images and an LSTM encoder for text, while the other employs CLIP encoders for both modalities. The text feature extraction is defined as

$$\text{TxE} = f_{\text{text}}(T), \quad (3)$$

where $T \in \mathbb{R}^{N \times d_t}$ represents the input text and f_{text} represents the text feature extraction function. N represents the batch size and d_t indicates the dimensionality of the text embeddings. Similarly, the image feature extraction is defined as

$$\text{IxE} = f_{\text{image}}(I), \quad (4)$$

where $I \in \mathbb{R}^{N \times H \times W \times 3}$ represents the input image tensor (with H and W representing the image height and width, and 3 indicating the RGB color channels) and f_{image} represents the image feature extraction function. This dual-path design enables flexible fusion strategies while ensuring consistency within modality-specific representations.

2.5. Feature fusion

Two feature fusion strategies are designed to enable multimodal fusion from image-only sources. They are based on textual descriptions automatically generated via AIDG, a ResNet+LSTM pipeline and an enhanced CLIP+PVD architecture. Both pipelines fuse visual features with AIDG-generated textual descriptions from the same image source. This enables multimodal classification without manual image descriptions.

In the ResNet+LSTM pipeline, image features are extracted using a pretrained ResNet50 model. The model outputs a 2048-dimensional

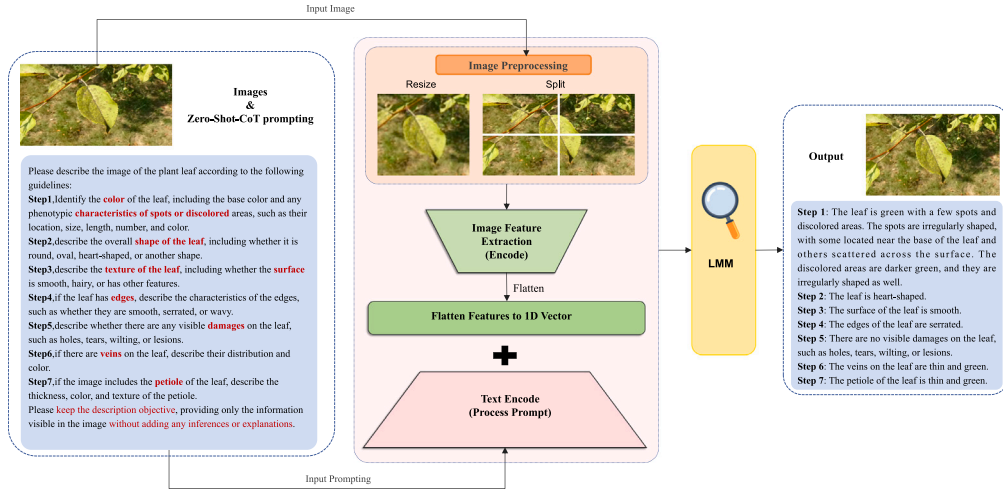


Fig. 2. Workflow of the proposed Automated Image Description Generation using LLMs (AIDG).

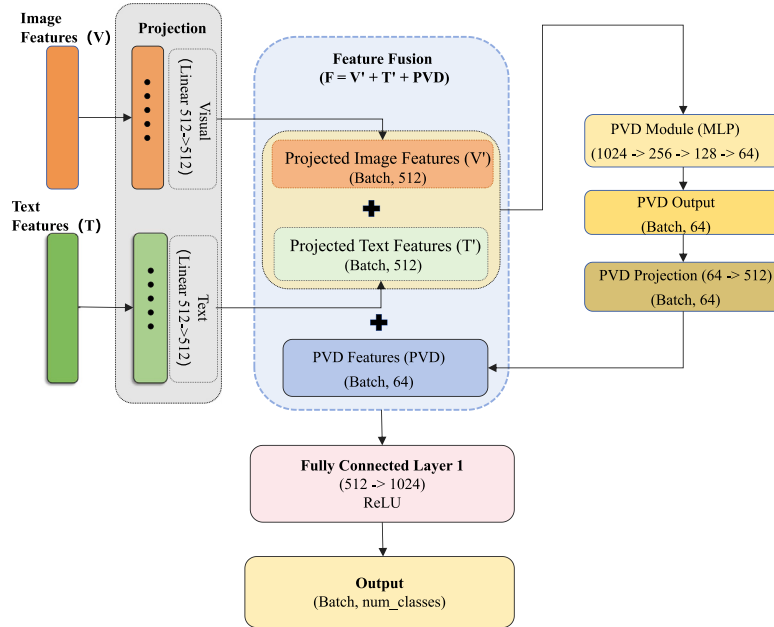


Fig. 3. Overview of the enhanced CLIP-based multimodal classification architecture with PVD integration. Visual and textual features are projected into a shared semantic space, concatenated, and further processed by the PVD module to generate a discriminative fused representation for classification.

global visual representation. Text sequences are first embedded through an embedding layer. They are then encoded using a single-layer LSTM network to obtain a fixed-length textual representation. The image and text features are concatenated and passed through two fully connected layers. These layers perform non-linear transformation and classification. The input visual features are $V \in \mathbb{R}^{N \times 2048}$, and the text features are $T \in \mathbb{R}^{N \times d_t}$. This architecture is facile and efficient, making it suitable for resource-constrained settings.

CLIP uses contrastive training, which does not explicitly model classification boundaries. Therefore, an enhanced CLIP-based architecture is proposed to address the existing limitations. This architecture incorporates a PVD module, as shown in Fig. 3. The visual and textual features extracted from the CLIP image and text encoders are projected into a unified 512-dimensional semantic space using linear projection functions

$$V' = f_{vp}(V), \quad T' = f_{tp}(T), \quad (5)$$

where $V \in \mathbb{R}^{N \times d_v}$, $T \in \mathbb{R}^{N \times d_t}$ represent the original CLIP image and text features, and $V' \in \mathbb{R}^{N \times 512}$, $T' \in \mathbb{R}^{N \times 512}$ represent the projected

features. These are concatenated and passed through a multi-layer perceptron to obtain a discriminative representation, expressed as

$$PVD = f_{pvd}([V', T']). \quad (6)$$

The final fused representation is formed by summing the projected features and the PVD output

$$F = V' + T' + PVD. \quad (7)$$

The fused vector is then passed through a classification head comprising a ReLU activation, dropout regularization, and two fully connected layers

$$X = \text{ReLU}(W_1 F + b_1), \quad (8)$$

$$X = \text{Dropout}(X), \quad (9)$$

$$Y = W_2 X + b_2, \quad (10)$$

where $W_1 \in \mathbb{R}^{512 \times 1024}$, $W_2 \in \mathbb{R}^{1024 \times C}$, and C indicate the number of output classes.

This fusion strategy preserves the strong alignment ability of CLIP and introduces explicit task-aware discriminative modeling using the PVD module. Consequently, it notably enhances the interaction between modalities and enhances classification performance. The improvement is particularly evident in fine-grained tasks, such as crop disease recognition. The method relies solely on visual input while automatically completing the multimodal structure. This demonstrates its practicality in data-scarce agricultural scenarios. This stage completes the second phase of the unified pipeline: combining AIDG-generated textual descriptions with visual features to enable multimodal classification.

2.6. Decision layer and loss function

The final *decision layer* of the model is implemented as a fully-connected (linear) layer that maps the fused feature representation to a C -dimensional vector of class logits, where C indicates the number of disease categories. Formally, the output of this layer is expressed as $Y \in \mathbb{R}^{N \times C}$, where N indicates the batch size.

During training, the model is optimized using the standard cross-entropy loss, which encourages the output logits to align with the ground-truth class labels. Given the predicted logits Y and the ground-truth label $y \in \{1, \dots, C\}$, the loss is computed as

$$\mathcal{L}_{CE} = - \sum_{i=1}^C \mathbb{1}(i = y) \log \left(\frac{\exp(Y_i)}{\sum_{j=1}^C \exp(Y_j)} \right). \quad (11)$$

This objective ensures that the model learns to assign high confidence scores to the correct class and is used consistently across all model variants for fair comparison.

3. Results and discussions

A series of experiments were conducted to evaluate the effectiveness and robustness of the proposed multimodal framework. The framework augments image-only samples with automatically generated textual descriptions to construct multimodal inputs. This section introduces the dataset, evaluation metrics, baseline models, and experimental configurations. Comparative analyses are performed to examine the impact of different feature encoders, text generation models, and fusion strategies. In addition, ablation studies are carried out to assess the individual contribution of the key component, the PVD module.

3.1. Dataset

The original PlantDoc dataset (Singh et al., 2020) was used without any manual relabeling or preprocessing. This dataset contains real-world images of crop diseases captured in agricultural fields. Unlike datasets collected in controlled environments, PlantDoc reflects the complexity of natural conditions. It includes variations in lighting, background clutter, and disease expression. These factors make it more challenging to develop models with strong robustness and generalization. Therefore, PlantDoc serves as an ideal benchmark for evaluating multimodal disease recognition models. The dataset includes images from 13 crop species and 18 disease types. We followed the original classification and split the data into training and test sets using an 80:20 ratio. No additional processing was applied to ensure that the models were evaluated on a clean and controlled benchmark. Notably, the original PlantDoc dataset provides only image data without paired textual descriptions. This limitation necessitates the use of automated text generation to construct multimodal inputs within the proposed framework.

3.2. Evaluation metrics

In this study, four primary evaluation metrics were employed: accuracy, precision, recall, and F1-score. These metrics comprehensively assess the performance of the models in processing multimodal data, particularly when dealing with crop disease images and text with significant background noise. They can effectively evaluate the classification accuracy and robustness of the models.

$$\text{Precision} = \frac{TP}{TP + FP} \times 100\%, \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \times 100\%, \quad (13)$$

$$\text{F1Score} = \frac{2TP}{2TP + FP + FN} \times 100\%, \quad (14)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%. \quad (15)$$

True Positives (TP) represent correctly identified positive samples. True Negatives (TN) represent correctly classified negative samples. False Positives (FP) represent negative instances that are mistakenly identified as positive. False Negatives (FN) represent positive instances that are incorrectly classified as negative. Precision measures the proportion of correctly predicted positive samples among all predicted positives. It reflects the classification accuracy for the positive class. Recall evaluates the proportion of actual positive samples that are correctly identified, reflecting the sensitivity of the model. The F1-score is the harmonic mean of precision and recall. It balances the two metrics to provide a comprehensive evaluation of model performance. Accuracy quantifies the overall proportion of correctly classified samples. It includes both positive and negative predictions.

3.3. Experimental settings

Fairness and reproducibility were ensured by conducting all experiments on a Linux server equipped with two NVIDIA GeForce RTX 3090 GPUs. A consistent data split strategy was adopted, with 80% of the data used for training and 20% for testing. From the training set, 10% was randomly selected as a validation set for model selection. The AdamW optimizer was used with an initial learning rate of $1e-4$. A StepLR learning rate scheduler was applied with a step size of 5 and gamma set to 0.5. Input images were resized to 224×224 and normalized using the mean and standard deviation of the ImageNet dataset. Data augmentation techniques such as random rotations, flipping, and cropping were applied to enhance generalization. All key hyperparameters (e.g., learning rate, weight decay, batch size, and dropout) were empirically tuned in preliminary experiments. These were adjusted within commonly reported ranges in the literature, rather than selected via grid search or Bayesian optimization. The final configuration was determined by balancing model performance and training cost using a practical empirical strategy. For CLIP-based models, additional metrics were recorded, including training time, peak GPU memory usage, and inference speed (FPS). All multimodal experiments were conducted using inputs derived solely from image-only data. In each case, AIDG was first applied to generate textual descriptions and construct multimodal input pairs. Experiments were conducted with five representative random seeds (42, 77, 0, 999, and 9999) to evaluate robustness and reproducibility under different initialization conditions. Seed 42 is generally used as a default in machine learning literature and serves as a baseline. Seed 77 is a mid-range prime number that provides good diversity. Seed 0 tests boundary initialization. Seeds 999 and 9999 are used to assess behavior in higher random spaces. This selection strategy covers low, medium, and high random seed spaces and strengthens the credibility of the stability analysis.

Table 1

Classification performance (mean $\pm$ standard deviation, %) of unimodal and multimodal models on crop disease recognition. All multimodal models were developed in this study. CLIP (ViT-L/14+PVD) achieves the best overall performance.

Modality	Model	Accuracy	Precision	Recall	F1
Unimodal	VGG16	53.73 $\pm$ 2.62	57.13 $\pm$ 3.52	53.73 $\pm$ 2.62	53.47 $\pm$ 3.23
	ViT	57.12 $\pm$ 0.92	60.16 $\pm$ 2.97	57.12 $\pm$ 0.92	56.99 $\pm$ 0.98
	ResNet50	55.85 $\pm$ 3.43	59.21 $\pm$ 3.72	55.85 $\pm$ 3.43	57.43 $\pm$ 2.59
	InceptionV3 (Yao et al., 2024)	51.99	–	–	48.33
	MobileNetV2 (Balafas et al., 2023)	57.20	–	–	–
	DenseNet121 (Balafas et al., 2023)	61.01	–	–	–
	GSMo-CNN (Yao et al., 2024)	63.30	–	–	63.76
Multimodal (Unimodal+Our AIDG)	VGG16 + RoBERTa	56.02 $\pm$ 1.39	60.08 $\pm$ 1.58	56.02 $\pm$ 1.39	55.96 $\pm$ 1.53
	ViT + RoBERTa	62.29 $\pm$ 3.0	65.6 $\pm$ 1.98	62.29 $\pm$ 3.0	64.13 $\pm$ 2.27
	ResNet50 + TF-IDF	59.07 $\pm$ 3.47	64.56 $\pm$ 4.31	59.07 $\pm$ 3.47	59.29 $\pm$ 3.7
	ResNet50 + LSTM	62.54 $\pm$ 2.72	65.92 $\pm$ 2.8	62.54 $\pm$ 2.72	62.6 $\pm$ 2.88
	CLIP (ViT-L/14 + PVD)	70.68 $\pm$ 1.93	73.79 $\pm$ 2.12	70.68 $\pm$ 1.93	70.76 $\pm$ 2.0

3.4. Baselines

Several representative models were selected as baselines to thoroughly validate the effectiveness of the proposed multimodal fusion framework. These baselines span different types of visual backbones and their combinations with textual representations. This setup enables a comprehensive evaluation across both unimodal and multimodal settings.

For the unimodal baselines, three widely adopted image classification architectures were employed: ResNet50, Vision Transformer (ViT) (Dosovitskiy, 2020), and VGG16 (Simonyan, 2014). ResNet50 is renowned for its deep residual structure and stable performance. ViT offers global modeling capability via self-attention mechanisms. VGG16 serves as a convolutional baseline. These models represent mainstream paradigms for image feature extraction. They enable us to benchmark the upper bound of vision-only approaches.

Additional baselines incorporating textual features were constructed to further assess multimodal integration. In particular, VGG16 and ViT were each combined with the RoBERTa language model to evaluate the role of textual semantics. All textual inputs in these baselines were automatically generated via AIDG from the corresponding image-only inputs. This design ensures consistency and scalability across all multimodal models. In addition, the combination of ResNet50 with TF-IDF, a classic sparse representation, was tested to evaluate multimodal performance under language modeling. These variants span a range of model capacities and complexities, forming a strong basis for evaluating the proposed method. Importantly, all textual features in these multimodal baselines were automatically generated via AIDG from image-only inputs. This enables model training and evaluation even when manually authored image–text pairs are not available.

3.5. Experimental results

Table 1 presents the classification performance of baseline architectures and the proposed multimodal fusion strategies for crop disease identification. Most multimodal models outperform their unimodal counterparts. This highlights the benefits of integrating visual and textual modalities. Importantly, all textual annotations were generated automatically by the AIDG module. This eliminates the need for manual labeling and enables multimodal deployment in image-only scenarios.

Among the unimodal baselines, lightweight convolutional networks such as MobileNetV2 and DenseNet121 achieved accuracy scores of 57.20% and 61.01%, respectively. Both outperformed VGG16 (accuracy: 53.73%, F1-score: 53.47%) and even deeper models such as ResNet50 (55.85%) and ViT (57.12%). These results suggest that compact architectures can deliver competitive performance in visual-only classification tasks. The GSMo-CNN model, based on the InceptionV3 backbone, achieved an F1-score of 63.76%. This substantially surpassed the plain InceptionV3 baseline (48.33%). The improvement is attributed to its multi-output structure, balance-weighted loss function,

and the use of transfer learning. This highlights the importance of backbone architecture and optimization strategies in unimodal settings.

In multimodal configurations, combining visual features with automatically generated textual descriptions consistently enhances performance. For example, the ResNet50+TF-IDF model exhibits a moderate improvement with an F1-score of 59.29%. Additionally, ViT+RoBERTa and ResNet50+LSTM achieved higher F1-scores of 64.13% and 62.60%, respectively. These results indicate that sequential modeling of generated descriptions can strengthen semantic alignment between visual and linguistic information, thereby improving classification accuracy.

Among all evaluated models, CLIP (ViT-L/14+PVD) achieved the best overall performance, with an accuracy of 70.68% and an F1-score of 70.76%. This demonstrates the strength of pretrained cross-modal embeddings in crop disease classification. By contrast, although VGG16+RoBERTa incorporates textual cues, its improvement is limited (F1-score: 55.96%). Its performance even lags behind the unimodal GSMo-CNN. This indicates that shallow visual backbones may lack the representational capacity required for effective multimodal integration.

We further substantiate the superiority of the proposed CLIP (ViT-L/14+PVD) architecture through pairwise *t*-tests against all baseline models using results from five independent runs with different random seeds. As indicated in Table 2, all comparisons yielded statistically significant differences ($p < 0.05$), indicating that the observed performance gains are unlikely to be caused by random fluctuations. In particular, the comparison with the ViT baseline resulted in a *t*-statistic of 15.484 and a *p* value of 1.02×10^{-4} , with a 95% confidence interval of [9.60, 13.80], confirming a marked, reliable improvement. Even when compared to strong multimodal baselines such as ResNet50+LSTM and VGG16+RoBERTa, the model consistently demonstrated significant advantages ($p < 0.005$). These results affirm the effectiveness and stability of the proposed multimodal fusion strategy.

All multimodal methods listed in Table 1 are implemented using the CogAgent-based AIDG module. In fact, the AIDG module supports various LMMs. The results of comparative LMM experiments have been presented in Section 3.6 to validate the performance differences of various LMMs within the proposed framework.

Fig. 4 displays the confusion matrix for the CLIP (ViT-L/14+PVD) model with a random seed of 42 to further elucidate the classification efficacy. The majority of classes are accurately classified, with misclassification predominantly occurring among visually analogous pathological conditions.

3.6. Comparison of text generation models: LLaVA vs CogAgent

To investigate the semantic differences in image–text generation across vision-language models and their impact on multimodal classification, we conducted a comparative study with two representative models: LLaVA v1.6 (Vicuna 7B) and CogAgent (cogvlm chat 17b). LLaVA has strong open instruction alignment capability. CogAgent performs well in multimodal understanding. Both support Zero-Shot

Table 2

Statistical significance test and confidence interval for F1 scores between the CLIP (ViT-L/14 + PVD) model and the baseline model.

Comparison	t-statistic	p-value	95% CI
vs VGG16	7.9704	1.34×10^{-3}	[9.23152, 19.10094]
vs ViT	15.484	1.02×10^{-4}	[9.60242, 13.79845]
vs ResNet50	8.6415	9.86×10^{-4}	[9.45394, 18.40478]
vs VGG16+Roberta	11.2035	3.61×10^{-4}	[10.72209, 17.78723]
vs ViT+Roberta	3.5368	2.41×10^{-2}	[1.13706, 9.44141]
vs ResNet50+TF-IDF	6.3741	3.11×10^{-3}	[6.19776, 15.76378]
vs ResNet50+LSTM	6.0666	3.73×10^{-3}	[3.22242, 8.66100]

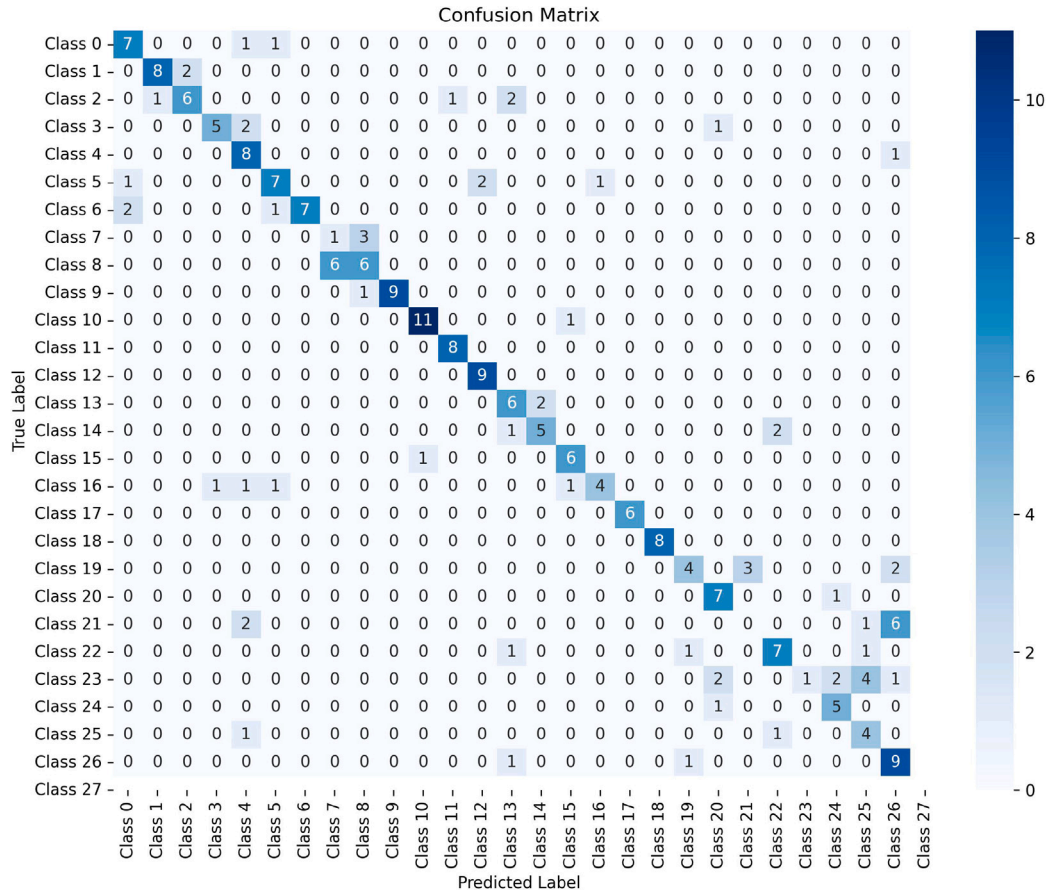


Fig. 4. Confusion matrix of the classification model across 28 leaf categories under random seed 42. Class indices correspond to the following categories: 0–Apple leaf, 1–Apple rust leaf, 2–Apple Scab Leaf, 3–Bell pepper leaf, 4–Bell pepper leaf spot, 5–Blueberry leaf, 6–Cherry leaf, 7–Corn Gray leaf spot, 8–Corn leaf blight, 9–Corn rust leaf, 10–Grape leaf, 11–Grape leaf black rot, 12–Peach leaf, 13–Potato leaf early blight, 14–Potato leaf late blight, 15–Raspberry leaf, 16–Soybean leaf, 17–Squash Powdery mildew leaf, 18–Strawberry leaf, 19–Tomato Early blight leaf, 20–Tomato leaf, 21–Tomato leaf bacterial spot, 22–Tomato leaf late blight, 23–Tomato leaf mosaic virus, 24–Tomato leaf yellow virus, 25–Tomato mold leaf, 26–Tomato Septoria leaf spot, 27–Tomato two spotted spider mites leaf.

image captioning; however, they differ significantly in visual encoders, pretraining objectives, and cross-modal alignment strategies.

Although both models are prompted with the same step-by-step instructions, their generated outputs show clear differences in structure and content. LLaVA closely follows the instructions and produces itemized descriptions in a “Step 1–Step 7” format. By contrast, CogAgent generates fluent, paragraph-style responses that combine multiple aspects into a single narrative. It often disregards the structural constraints specified in the prompt. These differences highlight the models’ varying capabilities in format control and response style. They are likely to result from differences in pretraining goals and fine-tuning strategies.

A unified Zero-Shot CoT prompt is applied to generate two sets of textual descriptions from the same PlantDoc images. One set is produced by LLaVA and the other by CogAgent. These textual descriptions are generated using AIDG and paired with the corresponding visual

features to develop two parallel sets of multimodal samples (Fig. 5). Each set is then fed into an identical multimodal classification model.

All other components are held constant, including the image encoder, text encoder, fusion module, and classifier head. This controlled setup enables us to isolate the effect of the language model. It provides a rigorous evaluation of how the semantic quality of generated texts influences overall multimodal classification performance.

This controlled comparison enables us to evaluate the influence of the language model alone, as all other factors are held constant. Moreover, it demonstrates that high-quality multimodal learning is achievable even in the absence of manually written image descriptions, as the entire image–text pairing process is automated.

The experimental results, as indicated in Table 3, reveal significant differences in multimodal classification performance between LLaVA and CogAgent. In the LLaVA model, the ResNet50+LSTM combination achieved the best performance, with an F1 score of 66.38%. This shows

Table 3
Comparison of multimodal classification performance (%) using textual descriptions generated by LLaVA and CogAgent.

Model	LLaVA				CogAgent			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
ResNet50 + TF-IDF	60.03	65.37	60.03	60.07	59.07	64.56	59.07	59.29
VGG16 + RoBERTa	55.93	61.05	55.93	55.98	56.02	60.08	56.02	55.96
ViT + RoBERTa	59.61	62.36	59.61	63.07	62.29	65.6	62.29	64.13
ResNet50 + LSTM	66.61	70.15	66.61	66.38	62.54	65.92	62.54	62.6
CLIP (ViT-L/14 + PVD)	67.03	71.55	68.72	65.29	70.68	73.79	70.68	70.76

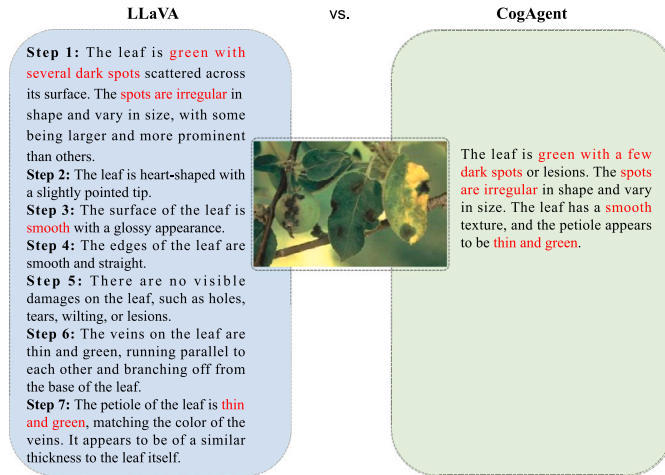


Fig. 5. Textual descriptions generated by LLaVA and CogAgent for the same image and prompt.

the effectiveness of combining a deep visual encoder such as ResNet50 with a sequence modeling LSTM. The combination captures complex relationships between image and text, which is particularly useful in disease recognition tasks.

By contrast, the CogAgent-based models, particularly CLIP (ViT-L/14) with the PVD module, achieved superior performance. The F1 score reached 70.76%. This result highlights the advantage of CLIP pretraining, which enhances visual-text alignment. The integration of the PVD module further enhanced both precision and recall.

Overall, LLaVA performs well in manually fused visual-text combinations. However, CogAgent achieves better results with pretrained CLIP architectures. Therefore, the choice of the text generation model should be based on the visual encoder and task requirements. In low-resource settings or in cases that require fine-grained reasoning, LLaVA with LSTM-based structured text generation may still provide practical advantages.

3.7. Comparison of feature encoders

Two mainstream model architectures were adopted to assess how different combinations of visual encoders and AIDG-generated textual features influence multimodal classification performance. The first uses traditional unimodal methods based on images. The second incorporates multimodal methods that combine images with descriptions. For image feature extraction, traditional visual encoders, such as ResNet50, ViT, and VGG16, were used. For the multimodal methods, AIDG-generated structured image descriptions (via LLaVA or CogAgent) were combined with corresponding visual encoders (e.g., ResNet50+LSTM, CLIP+ViT). This design enabled us to assess their performance in crop disease recognition tasks.

For each model, accuracy, precision, recall, and F1-score were calculated to evaluate classification performance under different task settings. Experimental results reveal that the model with LLaVA-generated image descriptions performed best in manually fused models (e.g.,

ResNet50+LSTM), achieving an F1-score of 66.38%. By contrast, models using the pretrained CLIP architecture combined with the PVD module (e.g., CLIP (ViT-L/14+PVD)) performed better when paired with CogAgent-generated image descriptions, reaching an F1-score of 70.76%.

A detailed comparison of model performance is reported in Tables 1 and 3. Among all the tested models, combining LLaVA-generated text with deep visual encoders (such as ResNet50+LSTM) effectively captured the complex relationship between image and text. This approach performed well in the crop disease recognition task.

This study confirms that classification performance can be greatly improved by combining structured image descriptions with deep visual features. The results highlight the critical role of the feature extraction module in enhancing robustness and accuracy. Moreover, they show that effective multimodal inputs can be constructed via automatic text generation by starting from image-only datasets. This strategy bridges the gap between real-world agricultural data and multimodal fusion frameworks.

3.8. Comparative analysis of CLIP variants with PVD

The influence of visual backbone configurations on both recognition performance and deployment efficiency was comprehensively evaluated by selecting three representative CLIP variants: ViT-B/32, ViT-B/16, and ViT-L/14. In this context, “ViT-B” and “ViT-L” denote base and large model scales, respectively. The numbers 32, 16, and 14 refer to patch sizes. Smaller patch sizes provide finer feature resolution and stronger representational power. However, they also increase the computational cost.

A comparison was conducted between the three CLIP backbone variants and their PVD-augmented versions to assess the impact of the PVD module on performance. Each model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score.

As shown in Fig. 6, the PVD module had varying effects on different CLIP backbones. For CLIP (ViT-B/32), although accuracy and other metrics show minor improvements, the F1-score decreases slightly. In particular, accuracy increases from 63.56% to 63.98%. Precision, recall, and F1-score display an overall upward trend. However, the F1-score remains slightly lower than the baseline (63.89% vs. 65.90%). A similar pattern is observed for CLIP (ViT-B/16). Accuracy increases from 67.80% to 69.07%, and the F1-score improves from 67.80% to 69.02%. These results indicate a positive impact from the PVD module, particularly in boosting recall and precision. By contrast, CLIP (ViT-L/14) shows the most substantial gains. Accuracy increases from 70.34% to 70.68%, and the F1-score improves from 67.99% to 70.76%. This indicates that the module helps higher-capacity models better integrate visual and textual information.

In terms of hardware efficiency, Fig. 7 shows that CLIP-ViT-B/32 achieves the shortest training time (60.47 min), lowest GPU memory consumption (1.07 GB), and highest inference speed (990.5 FPS). This makes it suitable for edge deployments. ViT-B/16 provides a balanced trade-off. ViT-L/14 achieves the best performance but incurs the highest computational cost.

As indicated in Table 4, different text generation models affect the final classification performance. The CogAgent-generated descriptions

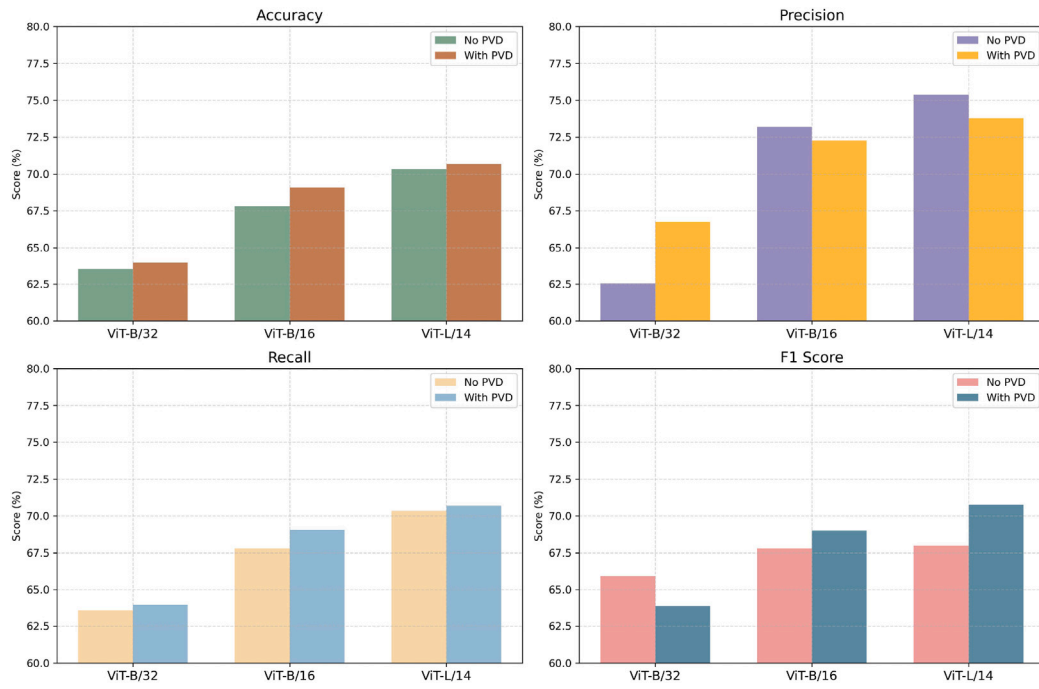


Fig. 6. Performance (%) comparison of different CLIP variants (ViT-B/32, ViT-B/16, ViT-L/14) with and without the PVD module across four metrics.

Table 4

Multimodal classification performance (%) of different CLIP variants under two text generation settings: LLaVA and CogAgent.

Model	LLaVA				CogAgent			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
CLIP (ViT-B/32)	61.95	64.25	61.95	62.99	63.56	62.55	63.56	65.90
CLIP (ViT-B/16)	65.43	69.85	65.43	64.94	67.80	73.21	67.80	67.80
CLIP (ViT-L/14)	66.27	73.51	66.27	63.12	70.34	75.37	70.34	67.99

Table 5

Ablation results showing performance (%) improvements from the discriminative projection (PVD) module across different visual backbones.

Model comparison	F1 score	Gain
CLIP (ViT-B/32) → CLIP (ViT-B/32)+PVD	65.90 → 63.89	-2.01
CLIP (ViT-B/16) → CLIP (ViT-B/16)+PVD	67.80 → 69.02	+1.22
CLIP (ViT-L/14) → CLIP (ViT-L/14)+PVD	67.99 → 70.76	+2.77

exhibit stronger alignment with CLIP architectures, particularly ViT-L/14, likely owing to their higher semantic precision. By contrast, LLaVA-generated descriptions exhibit stronger structural coherence, potentially supporting more effective multimodal alignment in certain architectures. This indicates that the choice of text generation strategy should be aligned with the design of the visual backbone in practical deployments.

These results indicate that the PVD module contributes to performance improvements across all tested CLIP variants, with the most noticeable benefits observed in larger-scale architectures. The module enhances the interaction between visual and textual modalities, which in turn strengthens the representational capacity and classification performance of multimodal systems. Moreover, selecting appropriate visual backbones and text generators is crucial for balancing performance and computational efficiency in real-world applications.

3.9. Ablation studies

All multimodal inputs in these ablation settings were constructed from image-only data using AIDG-generated textual descriptions. This

ensures full automation without human intervention. We further investigated the effectiveness of the proposed multimodal framework by conducting ablation experiments to examine the contribution of the PVD module across different CLIP-based visual backbones.

As indicated in Table 5, adding PVD to a CLIP (ViT-L/14) architecture trained on CogAgent-generated image descriptions significantly improves the F1 score. The score increases from 67.99% to 70.76%. This gain demonstrates that the discriminative projection mechanism enhances modality interaction and strengthens the alignment between visual and textual representations.

However, for CLIP (ViT-B/32), the F1 score slightly decreases (from 65.90% to 63.89%) after adding the PVD module. This indicates that the benefit of discriminative projection may depend on the representational capacity of the visual backbone. By contrast, consistent improvements are observed for CLIP (ViT-B/16) and CLIP (ViT-L/14), with F1 score gains of +1.22 and +3.43, respectively.

These results highlight the practical value of the PVD module in improving classification performance, particularly when paired with stronger visual backbones. Moreover, the results validate its effectiveness as a structural enhancement module within the multimodal recognition framework.

3.10. Discussions

The experimental results presented in Section 3 demonstrate the effectiveness and robustness of the proposed multimodal framework. The framework enhances image-only crop disease samples by incorporating automatically generated textual descriptions. These descriptions create semantically enriched inputs for classification. In this section, key findings are summarized and analyzed.

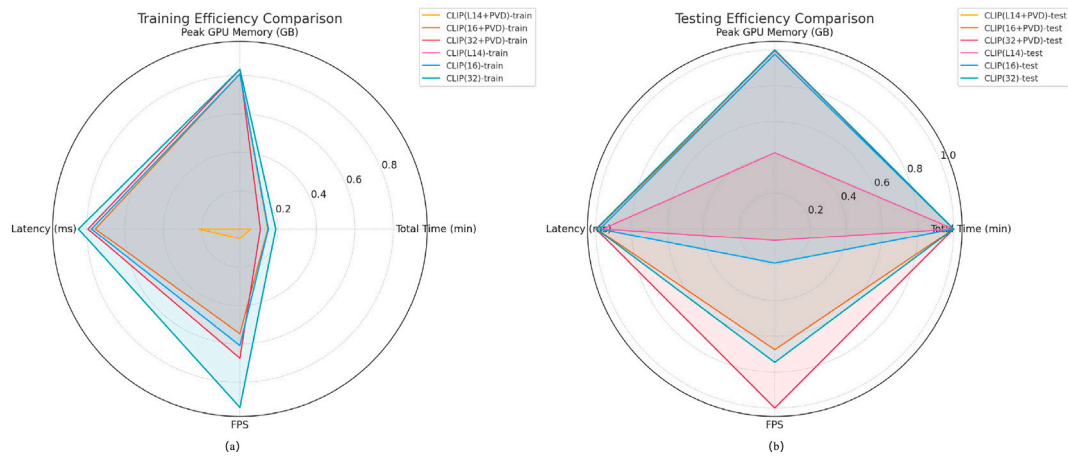


Fig. 7. (a) Normalized radar chart of training efficiency metrics for CLIP variants, including total training time, peak GPU memory usage, inference latency, and average FPS. (b) Inference efficiency comparison across CLIP variants. Models with PVD (e.g., ViT-B/32) achieve favorable trade-offs between performance and efficiency.

First, the core hypothesis has been validated. In particular, fusing vision and language modalities, particularly through AIDG-generated descriptions, enhances classification performance. For instance, the unimodal ResNet50 baseline yielded an F1-score of 57.43%. This score increased to 62.6% after CogAgent-generated descriptions were integrated, as indicated in Table 1. Similarly, the CLIP-based architecture, augmented with the PVD module and CogAgent-generated text, achieved an F1-score of 70.76%. The performance was further confirmed through statistical analysis (Table 2). These consistent improvements indicate that structured textual information provides complementary semantic cues. Such cues help resolve visually subtle or occluded disease symptoms.

Second, the comparison between CogAgent and LLaVA text generators revealed clear differences in cross-modal alignment. Descriptions generated by LLaVA follow a structured, step-wise format. They align more effectively with sequential encoders such as LSTM. By contrast, CogAgent produces fluent, paragraph-style outputs that better align with global semantic embedding mechanisms used in CLIP. This explains the superior performance of LLaVA in the ResNet50+LSTM pipeline (F1: 66.38%). It also explains the advantage of CogAgent in the CLIP+PVD configuration (F1: 70.76%). These findings underscore the importance of modality compatibility in designing multimodal systems.

Third, the incorporation of the PVD module results in significant performance gains in high-capacity backbones. As reported in Table 5, the ViT-L/14 variant showed a 2.77% improvement in F1-score following PVD integration. This indicates that the PVD module enhances multimodal interaction by projecting visual and textual features into a shared discriminative space. Consequently, semantic alignment is strengthened and the modality gap is reduced. However, smaller architectures, such as ViT-B/32, exhibited slight performance degradation. This implies that the benefits of PVD may be contingent upon sufficient representational capacity. Fig. 7 illustrates the efficiency trade-offs among different backbone configurations.

Fourth, the confusion matrix presented in Fig. 4 shows recurring misclassifications among visually similar tomato diseases. In particular, *Tomato leaf bacterial spot* (class 21), *Tomato Septoria leaf spot* (class 26), and *Tomato Early blight leaf* (class 19) are often misclassified as *Tomato leaf yellow virus* (class 24). These diseases exhibit overlapping lesion patterns and similar textures, which makes them harder to distinguish based on global image-level features alone. Future work should incorporate lesion-level segmentation or domain-specific visual priors. These additions could improve fine-grained discrimination, particularly for visually similar symptom classes.

Fifth, the PlantDoc dataset reflects realistic agricultural conditions, such as natural lighting and background clutter. However, it also

has inherent limitations, including class imbalance, lack of temporal progression, and absence of auxiliary sensor information. These problems may hinder generalization under diverse field scenarios. The lack of lesion-level annotations further restricts interpretable and spatially attentive classification. Informal observations also indicate that the AIDG module remains relatively robust under variations in crop type, lighting, and image quality. A formal robustness evaluation is planned for future work.

Finally, several methodological advantages emerge when compared with existing approaches. For example, Radford et al. (2021) proposed CLIP for vision-language alignment but required domain-specific fine-tuning. By contrast, the current framework adapts CLIP through structured Zero-Shot prompting, without manual annotation. Similarly, Lu et al. (2024) presented a Transformer-based fusion method that relies on manually written descriptions, which limits scalability. This limitation is addressed by the automatic description generation of AIDG. Furthermore, Garg et al. (2021) focused on Internet-of-Things and sensor integration. By contrast, the proposed method enhances visual representation directly through language, offering a complementary modality. These comparisons highlight the improved scalability and automation of the framework relative to manually supervised or sensor-dependent systems.

In addition to quantitative results, a potential deployment of the proposed framework in real agricultural settings is illustrated. Fig. 8 presents a potential implementation scenario in farming operations. Crop images are acquired in the field using unmanned aerial vehicles or mobile devices. These images are then processed by the multimodal plant disease recognition system. The outputs can be applied to multiple tasks, such as disease diagnostics, agronomic advisory services, early warning, and market analysis. This scenario highlights the scalability and practical value of the framework beyond experimental settings. It also demonstrates the potential for integration into precision agriculture and digital farming practices.

The proposed AIDG-based multimodal framework improves classification accuracy by enriching visual inputs with structured semantic descriptions. The results highlight the importance of appropriately aligning visual and textual components. Moreover, they suggest future directions, including enhanced fine-grained reasoning, better spatial focus, and stronger generalization across varied agricultural conditions.

4. Conclusions

This study presents an automated multimodal framework for crop disease recognition. The proposed framework combined visual and textual information to enhance classification performance under field

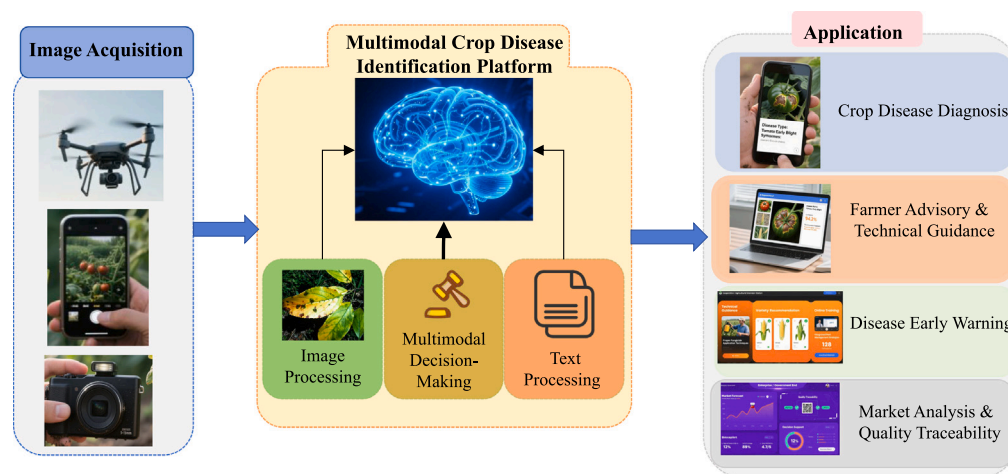


Fig. 8. Application scenario of the proposed multimodal crop disease recognition framework in real agricultural settings.

conditions. Large vision-language models were employed to generate structured textual descriptions of disease symptoms. These descriptions enriched image data with semantic context. The textual information was then fused with visual features via tailored multimodal classification pipelines.

Experimental evaluations on the PlantDoc dataset showed that integrating generated text enhanced model robustness and accuracy across different backbones. The ResNet50+LSTM classifier paired with LLaVA-generated descriptions outperformed unimodal baselines. Similarly, the CLIP (ViT-L/14) model combined with CogAgent descriptions and a discriminative PVD module achieved superior results. Both configurations surpassed unimodal approaches by a notable margin. These results confirmed the utility of textual augmentation and modality-aware fusion strategies in complex agricultural scenarios.

While the framework performed well overall, certain challenges remain. Misclassification persists among diseases with similar visual traits, such as leaf spot variants and viral infections. This indicates the need for more localized or fine-grained modeling. The computational cost also varies across model configurations. This variation may affect deployment feasibility in resource-limited environments.

Future work will explore lesion segmentation to enhance local symptom representation. Moreover, we plan to expand the framework to broader crop categories and geographic domains. Furthermore, additional modalities, such as environmental sensors or hyperspectral imaging, will be integrated. These enhancements will aim to further improve classification accuracy, generalizability, and practical applicability in precision agriculture.

CRedit authorship contribution statement

Shufeng Xiong: Supervision, Resources, Project administration, Methodology. **Lanlan Wang:** Writing – review & editing, Writing – original draft, Methodology. **Yiming Zhang:** Software, Methodology. **Ping Dong:** Resources, Data curation. **Bingkun Wang:** Supervision, Funding acquisition, Conceptualization. **Yinchao Che:** Validation, Investigation. **Lei Shi:** Visualization, Validation, Conceptualization. **Haiping Si:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code and data availability

The source code has already been made publicly available at <https://github.com/sylvia07/AIDG.git>.

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